The Mollified Polygon Overlap: Area, Gradient, and Exact Newton Minimization

Derivation notes

Abstract

A self-contained derivation of the Plummer-mollified overlap area between two convex polygons, its exact vertex gradient, and the Newton scheme they enable. Everything reduces, in closed form, to elementary functions plus a *single* transcendental — the Clausen function Cl_2 — and the whole evaluation is carried out without complex arithmetic. Numerical verification of every step is collected in §11.

Contents

1 Why mollify

Let A and B be simple polygons with CCW vertices $\vec{v}_k$ and edges $\vec{e}_k = \vec{v}_{z(k)} - \vec{v}_k$ ($z(k)$ the next vertex). The sharp overlap area

$$A_\cap = \int_{\mathbb{R}^2} \mathbf{1}_A(\vec{x})\, \mathbf{1}_B(\vec{x})\, d^2x \tag{1.1}$$

is a perfectly good energy, but its *gradient* is only C^0: as a vertex of A slides across an edge of B, the contact set changes non-smoothly and $\partial A_\cap/\partial \vec{v}$ has a kink. Two consequences:

- first-order relaxation (FIRE) stalls — in practice $\max|F|$ floors around 3.6×10^{-3};
- the Hessian is undefined at contacts, so no second-order method can be used at all.

Both problems disappear if we smooth the indicator. Convolving *one* of the two indicators with a mollifier K_σ gives

$$A_\cap^\sigma = \int_{\mathbb{R}^2} \mathbf{1}_A\,(K_\sigma * \mathbf{1}_B)\, d^2x = \iint d^2x\, d^2x'\; \mathbf{1}_A(\vec{x})\, K_\sigma(\vec{x}-\vec{x}')\, \mathbf{1}_B(\vec{x}'), \tag{1.2}$$

which is C^∞ in the vertices for any smooth K_σ and converges to (1.1) as $\sigma \to 0$. The parameter σ is a genuine model choice (a softening length), not a numerical fudge.

Choice of kernel. A Gaussian makes the boundary integrals intractable. The 2D Plummer profile

$$K_\sigma(\vec{r}) = \frac{\sigma^2}{\pi\,(\sigma^2 + |\vec{r}|^2)^2}, \qquad \int_{\mathbb{R}^2} K_\sigma = 1, \tag{1.3}$$

is the right compromise: it is smooth, normalized, and — as we now show — its potential is a logarithm, which is exactly what makes every edge-pair integral doable.

2 From a double area integral to a double boundary integral

2.1 The field and the potential

Seek $\vec{E}_\sigma$ with $\nabla\cdot\vec{E}_\sigma = K_\sigma$. Try $\vec{E}_\sigma = C\vec{r}/(\sigma^2+r^2)$:

$$\frac{\partial}{\partial x}\frac{Cx}{\sigma^2+r^2} = C\,\frac{\sigma^2+y^2-x^2}{(\sigma^2+r^2)^2},$$

so by symmetry

$$\nabla\cdot\vec{E}_\sigma = C\,\frac{(\sigma^2+y^2-x^2)+(\sigma^2+x^2-y^2)}{(\sigma^2+r^2)^2} = 2\pi C\,\frac{\sigma^2}{\pi(\sigma^2+r^2)^2} = 2\pi C\,K_\sigma.$$

Matching forces $C = 1/2\pi$:

$$\vec{E}_\sigma(\vec{r}) = \frac{\vec{r}}{2\pi(|\vec{r}|^2+\sigma^2)}. \tag{2.1}$$

Next seek Φ_σ with $\nabla\Phi_\sigma = \vec{E}_\sigma$. Trying $\Phi_\sigma = C\ln(r^2+\sigma^2)$ gives $\nabla\Phi_\sigma = 2C\,\vec{r}/(r^2+\sigma^2)$, so $2C = 1/2\pi$ and

$$\boxed{\Phi_\sigma(r) = \frac{1}{4\pi}\ln(r^2+\sigma^2)} \tag{2.2}$$

2.2 Two applications of the divergence theorem

Define the *softened membership function*

$$\Psi_B^\sigma(\vec{x}) \equiv (K_\sigma * \mathbf{1}_B)(\vec{x}) = \int d^2y\,(\nabla\cdot\vec{E}_\sigma)(\vec{y}-\vec{x})\,\mathbf{1}_B(\vec{y}). \tag{2.3}$$

Gauss's theorem on B turns the area integral into a boundary integral,

$$\Psi_B^\sigma(\vec{x}) = \oint_{\partial B}\vec{E}_\sigma(\vec{y}-\vec{x})\cdot\hat{n}_B(\vec{y})\,d\ell_y. \tag{2.4}$$

As $\sigma\to 0$, $\Psi_B^\sigma \to 1$ strictly inside B, $\to \frac{1}{2}$ on an edge, $\to$ (interior angle)$/2\pi$ at a vertex, and $\to 0$ outside: it is a smoothed indicator, which is the useful way to think about it.

Now $A_\cap^\sigma = \int_A \Psi_B^\sigma\,d^2x$. Writing $\vec{E}_\sigma = \nabla_{\vec{x}}\Phi_\sigma$ and applying the gradient theorem $\int_\Omega \nabla u\,dV = \oint_{\partial\Omega} u\,\hat{\nu}\,dS$ on A,

$$\boxed{A_\cap^\sigma = -\oint_{\partial A} d\ell_x \oint_{\partial B} d\ell_y\;\Phi_\sigma(|\vec{x}-\vec{y}|)\;\hat{n}_A(\vec{x})\cdot\hat{n}_B(\vec{y})} \tag{2.5}$$

The two-dimensional overlap has become a *one*-dimensional double integral over the two boundaries. With (2.2) and polygonal boundaries this is a finite sum over edge pairs:

$$A_\cap^\sigma = -\frac{1}{4\pi}\sum_{\alpha\in\partial A}\sum_{\beta\in\partial B}(\hat{n}_\alpha\cdot\hat{n}_\beta)\int_{e_\alpha} d\ell_x \int_{e_\beta} d\ell_y\,\ln\big(|\vec{x}-\vec{y}|^2+\sigma^2\big). \tag{2.6}$$

Everything that follows is the evaluation of one such edge-pair panel.

3 The edge-pair panel

3.1 The normalized frame

Everything that follows is one edge of ∂A against one edge of ∂B, so fix the frame once, at the start, and never change variables again. Parametrize

$$\vec{x} = \vec{v}_\alpha + t\,\vec{e}_\alpha, \qquad \vec{y} = \vec{v}_\beta + s\,\vec{e}_\beta, \qquad t, s \in [0,1],$$

write $L \equiv |\vec{e}_\alpha|$ and $\vec{\Delta} \equiv \vec{v}_\alpha - \vec{v}_\beta$, and rotate so that $\vec{e}_\alpha = (L, 0)$. Then

$$\vec{x} - \vec{y} = \big(\ \Delta^x - e^x_\beta s + Lt\ ,\ \ \Delta^y - e^y_\beta s\ \big), \tag{3.1}$$

whose second component carries *no* t: moving along the inner edge changes only the coordinate parallel to it. That is the one structural fact the whole reduction rests on. Measure the transverse component in units of σ and name, once, the variable X and the three parameters m, ν, w that carry the rest of these notes,

$$\boxed{X \equiv \frac{\Delta^y - e^y_\beta s}{\sigma}, \qquad m \equiv \frac{e^x_\beta}{e^y_\beta}, \qquad \nu_0 \equiv \frac{\Delta^x - m\Delta^y}{\sigma}, \quad \nu_1 \equiv \nu_0 + \frac{L}{\sigma}, \qquad w \equiv \sqrt{1 + m^2 + \nu^2}} \tag{3.2}$$

with ν standing for either ν_0 or ν_1. Eliminating s in favor of X turns the parallel component of (3.1) into $\sigma(\nu_0 + mX)$ at $t = 0$ and $\sigma(\nu_1 + mX)$ at $t = 1$: m is the slope of the outer edge in this frame, and ν_0, ν_1 are the scaled parallel offsets of the inner edge's two endpoints. So the softened square distance at either endpoint is $\sigma^2 F(X)$ with

$$F(X) \equiv (\nu + mX)^2 + X^2 + 1 = AX^2 + BX + C, \qquad A = 1 + m^2, \quad B = 2m\nu, \quad C = 1 + \nu^2. \tag{3.3}$$

Completing the square in (3.3) produces w and, with it, the identity that removes every branch condition and all case analysis from these notes:

$$A\,F(X) = (AX + m\nu)^2 + w^2, \qquad w^2 = 1 + m^2 + \nu^2 > 0 \quad \text{always} \qquad \big(\text{equivalently } 4AC - B^2 = 4w^2\big). \tag{3.4}$$

So F has *no real roots* for any m, ν: every antiderivative below is a real logarithm plus a real arctangent, and the dilogarithm that eventually appears is genuinely irreducible (§5).

3.2 The panel

The inner integral $\int_0^1 \ln(|\vec{x} - \vec{y}|^2 + \sigma^2)\,dt$ is now a single antiderivative. By (3.1) the transverse part is frozen at $\rho \equiv \sigma\sqrt{X^2 + 1}$ while the parallel part sweeps $u = \sigma(\nu_0 + mX) \to \sigma(\nu_1 + mX)$, so with $du = L\,dt$ and $\int \ln(u^2 + \rho^2)du = u\ln(u^2 + \rho^2) - 2u + 2\rho \arctan\frac{u}{\rho}$,

$$\boxed{\int_0^1 \ln\big(|\vec{x} - \vec{y}|^2 + \sigma^2\big)dt = \frac{\sigma}{L}\Big[\,(\nu + mX)\ln\big(\sigma^2 F(X)\big) - 2(\nu + mX) + 2\sqrt{X^2 + 1}\ \Theta(X)\,\Big]_{\nu=\nu_0}^{\nu=\nu_1}} \tag{3.5}$$

because $u^2 + \rho^2 = \sigma^2 F(X)$ and $u/\rho = (\nu + mX)/\sqrt{X^2 + 1}$ at both endpoints. That last ratio is the object the rest of these notes is about:

$$\boxed{\Theta(X) \equiv \arctan\frac{\nu + mX}{\sqrt{X^2 + 1}}} \tag{3.6}$$

No auxiliary quadratic coefficients, no discriminant, and no separate arctangent difference: the two ν values *are* the two ends of the inner edge, and Θ emerges on its own.

3.3 Outer integral

Integrate (3.5) over s using $ds = -\frac{\sigma}{e^y_\beta}dX$, with endpoints $X_0 = \Delta^y/\sigma$ and $X_1 = (\Delta^y - e^y_\beta)/\sigma$. The middle term of (3.5) is a polynomial and the $\ln\sigma^2$ splits off, leaving exactly two nonelementary-looking integrals:

$$\frac{I_2}{|\vec{e}_\beta|} = -\frac{\sigma^2}{L\,e^y_\beta}\Bigg[\underbrace{\int_{X_0}^{X_1}(\nu+mX)\ln F\,dX}_{\text{elementary}} + 2\underbrace{\int_{X_0}^{X_1}\sqrt{X^2+1}\,\Theta\,dX}_{\text{the only hard object}} + \text{polynomial in } X\Bigg]_{\nu=\nu_0}^{\nu=\nu_1}, \tag{3.7}$$

where the two polynomial terms of (3.5) share a coefficient and collect as

$$\Big[\nu(X_1{-}X_0) + \tfrac{m}{2}(X_1^2{-}X_0^2)\Big]\big(\ln\sigma^2 - 2\big).$$

Its -2 half reproduces the bare constant -2 of the old four-piece form, since $\nu_1 - \nu_0 = L/\sigma$ and $X_1 - X_0 = -e^y_\beta/\sigma$.

The logarithmic piece. It is elementary: with $\Lambda_0 = \int \ln F\,dX$ and $\Lambda_1 = \int X\ln F\,dX$,

$$\Lambda_0(X) = \Big(X + \tfrac{B}{2A}\Big)\ln F - 2X + \frac{2w}{A}\arctan\frac{AX+m\nu}{w}, \tag{3.8}$$

$$\Lambda_1(X) = \Big(\tfrac{X^2}{2} - \tfrac{B^2-2AC}{4A^2}\Big)\ln F - \tfrac{X^2}{2} + \tfrac{BX}{2A} - \frac{Bw}{A^2}\arctan\frac{AX+m\nu}{w}, \tag{3.9}$$

giving $\int_{X_0}^{X_1}(\nu{+}mX)\ln F\,dX = \big[\nu\Lambda_0{+}m\Lambda_1\big]_{X_0}^{X_1}$, which closes the logarithmic half of (3.7) completely.

The arctangent piece. Everything hard in this problem is now the one integral $\int\sqrt{X^2+1}\,\Theta\,dX$ — and, as §4 shows, the two cousins of it that the gradient needs.

4 The master arctangent integral

The energy needs $\int\sqrt{X^2+1}\,\Theta\,dX$; the gradient (§6) will need $\int\frac{X}{\sqrt{X^2+1}}\Theta\,dX$ and $\int\frac{X^2}{\sqrt{X^2+1}}\Theta\,dX$. All three are instances of one object. Define, for any weight V',

$$\boxed{\mathbb{M}[V] \equiv \int V'(X)\,\Theta(X)\,dX = V(X)\Theta(X) - \int V(X)\,\Theta'(X)\,dX} \tag{4.1}$$

The derivative of Θ is the reason this works. Writing $\tau = (\nu+mX)/\sqrt{X^2+1}$ for its argument,

$$\tau' = \frac{m(X^2+1) - X(\nu+mX)}{(X^2+1)^{3/2}} = \frac{m-\nu X}{(X^2+1)^{3/2}}, \qquad 1+\tau^2 = \frac{F(X)}{X^2+1},$$

the second identity being (3.3) itself, so

$$\Theta'(X) = \frac{\tau'}{1+\tau^2} = \frac{m-\nu X}{\sqrt{X^2+1}\,F(X)}. \tag{4.2}$$

The $\sqrt{X^2+1}$ in the denominator is what makes the three weights close on the same objects. Introduce

$$v_\pm(X) \equiv \tfrac{1}{2}\Big(X\sqrt{X^2+1} \pm \operatorname{arcsinh} X\Big), \qquad v_+' = \sqrt{X^2+1}, \quad v_-' = \frac{X^2}{\sqrt{X^2+1}}, \tag{4.3}$$

(check: $\frac{d}{dX}\,X\sqrt{X^2+1} = \frac{2X^2+1}{\sqrt{X^2+1}}$ and $\frac{d}{dX}\,\operatorname{arcsinh} X = \frac{1}{\sqrt{X^2+1}}$), and the two elementary primitives

$$\lambda(X) \equiv \int \frac{m-\nu X}{F}\,dX, \qquad \varrho(X) \equiv \int \frac{X(m-\nu X)}{F}\,dX, \tag{4.4}$$

and the one transcendental

$$\boxed{J_{\operatorname{arcsinh}} \equiv \tfrac{1}{2}\int \frac{\operatorname{arcsinh} X\,(\nu X - m)}{\sqrt{X^2+1}\,F(X)}\,dX} \tag{4.5}$$

Then, splitting $v_\pm$ into its $X\sqrt{X^2+1}$ and arcsinh halves against (4.2) (the first half cancels the $\sqrt{X^2+1}$ and leaves ϱ; the second leaves $\mp 2J_{\operatorname{arcsinh}}$):

weight V'	V	$\mathbb{M}[V]$	transcendental	used by
$\sqrt{X^2+1}$	v_+	$v_+\Theta - \frac{1}{2}\varrho + J_{\operatorname{arcsinh}}$	$+J_{\operatorname{arcsinh}}$	energy (3.7)
$X/\sqrt{X^2+1}$	$\sqrt{X^2+1}$	$\sqrt{X^2+1}\,\Theta - \lambda$	none	gradient W^0
$X^2/\sqrt{X^2+1}$	v_-	$v_-\Theta - \frac{1}{2}\varrho - J_{\operatorname{arcsinh}}$	$-J_{\operatorname{arcsinh}}$	gradient W^1

(4.6)

Three remarks, all useful later:

- The middle row is *purely elementary* — the $\sqrt{X^2+1}$ weight cancels the one in Θ' outright. The gradient's W^0 costs no special functions at all.
- The outer rows differ only through $v_\pm = \frac{1}{2}(X\sqrt{X^2+1} \pm \operatorname{arcsinh} X)$, so they carry *the same* $J_{\operatorname{arcsinh}}$ with opposite sign. Energy and force share one transcendental.
- Consequently $\mathbb{M}[\sqrt{\cdot}]+\mathbb{M}[X^2/\sqrt{\cdot}] = X\sqrt{X^2+1}\,\Theta - \varrho$ is elementary (the dilogarithms cancel), a cheap internal consistency check.

The elementary primitives explicitly. Using $4AC - B^2 = 4w^2$ and $\int \frac{dX}{F} = \frac{1}{w}\arctan\frac{AX+m\nu}{w}$:

$$\lambda(X) = -\frac{\nu}{2A}\ln F + \frac{mw}{A}\arctan\frac{AX+m\nu}{w}, \tag{4.7}$$

$$\varrho(X) = -\frac{\nu X}{A} + \frac{P}{2A}\ln F + \Big(Q - \frac{PB}{2A}\Big)\frac{1}{w}\arctan\frac{AX+m\nu}{w}, \qquad P = m + \frac{\nu B}{A},\quad Q = \frac{\nu C}{A}. \tag{4.8}$$

(For (4.7): write $m - \nu X = -\frac{\nu}{2A}(2AX+B) + \big(m + \frac{\nu B}{2A}\big)$ and note $m + \frac{\nu B}{2A} = \frac{mw^2}{A}$.)

5 Evaluating $J_{\operatorname{arcsinh}}$: a fully real route

$J_{\operatorname{arcsinh}}$ is genuinely dilogarithm-level: by (3.4) the quadratic F always has a complex-conjugate root pair, so no algebraic manipulation reduces it to elementary functions. What we *can* do is evaluate it without ever performing complex arithmetic. That is this section.

5.1 Setup and integration by parts

Substitute $X = \sinh t$ (so $\operatorname{arcsinh} X = t$, $dX/\sqrt{X^2+1} = dt$), with $t_0 = \operatorname{arcsinh} X_0$, $t_1 = \operatorname{arcsinh} X_1$:

$$J_{\operatorname{arcsinh}} = \tfrac{1}{2}\int_{t_0}^{t_1} t\,\frac{\nu\sinh t - m}{Q(t)}\,dt, \qquad Q(t) \equiv A\sinh^2 t + B\sinh t + C. \tag{5.1}$$

Integrate by parts with $u = t$, $dv = (\nu \sinh t - m)\, dt/Q$, i.e. $V(t) \equiv \int (\nu \sinh t - m)\, dt/Q$:

$$J_{\text{arcsinh}} = \tfrac{1}{2}\big[t\, V(t)\big]_{t_0}^{t_1} - \tfrac{1}{2} \int_{t_0}^{t_1} V(t)\, dt. \tag{5.2}$$

Both pieces will turn out real, and V will turn out to be a difference of two arctangents.

5.2 The rational factor, its poles and residues

Put $y = e^t$, so $\sinh t = \frac{1}{2}(y - 1/y)$ and $dt = dy/y$. Then $\nu \sinh t - m = N(y)/(2y)$ and $Q = D(y)/(4y^2)$ with

$$\begin{gathered} \frac{\nu \sinh t - m}{Q}\, dt = 2R(y)\, dy, \qquad R(y) \equiv \frac{N(y)}{D(y)}, \\ N(y) = \nu(y^2 - 1) - 2my, \qquad D(y) = A(y^4 + 1) + 2B(y^3 - y) + (4C - 2A)y^2. \end{gathered} \tag{5.3}$$

so $V = \int 2R\, dy$. The poles of R are the roots of D. Now $Q = 0$ means $\sinh t = \zeta$ with ζ a root of $A\zeta^2 + B\zeta + C$; by (3.4),

$$\zeta = \frac{-m\nu \pm iw}{1 + m^2}. \tag{5.4}$$

Each $\sinh t = \zeta$ is $y^2 - 2\zeta y - 1 = 0$, i.e. $y = \zeta \pm \sqrt{\zeta^2 + 1}$, so D has four roots. For the upper ζ one finds $\zeta^2 + 1 = (mw - i\nu)^2/(1 + m^2)^2$, hence $\sqrt{\zeta^2 + 1} = (mw - i\nu)/(1 + m^2)$, and $\zeta \pm \sqrt{\zeta^2 + 1}$ collapses — with *no complex square root left* — to the two poles in the upper half-plane,

$$\boxed{\eta_+ = \frac{(w - \nu)(m + i)}{1 + m^2}, \qquad \eta_- = \frac{(w + \nu)(-m + i)}{1 + m^2}} \tag{5.5}$$

the other two being the conjugates $\bar{\eta}_\pm$. Their real coordinates are

$$a_\pm \equiv \operatorname{Re} \eta_\pm = \frac{\pm m(w \mp \nu)}{1 + m^2}, \qquad b_\pm \equiv \operatorname{Im} \eta_\pm = \frac{w \mp \nu}{1 + m^2}\ (> 0). \tag{5.6}$$

Note for later that $\eta_+\eta_- = -1$, since $(m + i)(-m + i) = -(1 + m^2)$ and $(w - \nu)(w + \nu) = 1 + m^2$.

Residues. The residue of R at a simple pole η is $c = N(\eta)/D'(\eta)$. On the factored denominator $D = A(y^2 - 2\zeta_1 y - 1)(y^2 - 2\zeta_2 y - 1)$ a pole $\eta = \zeta_i \pm r_i$ ($r_i = \sqrt{\zeta_i^2 + 1}$) obeys $\eta^2 = 2\zeta_i \eta + 1$, so $N(\eta) = 2\eta(\nu\zeta_i - m)$ and $D'(\eta) = 4A(\pm r_i)(\zeta_i - \zeta_j)\eta$, giving

$$c = \frac{N(\eta)}{D'(\eta)} = \frac{\nu\zeta_i - m}{2A(\pm r_i)(\zeta_i - \zeta_j)} = \pm\frac{i}{4}, \tag{5.7}$$

purely imaginary, because $\nu\zeta_i - m = -w\, r_i$ (the $mw - i\nu$ of r_i cancels the numerator) and $\zeta_1 - \zeta_2 = 2iw/(1 + m^2)$. Tracking the branch, $c(\eta_+) = +\frac{i}{4}$, $c(\eta_-) = -\frac{i}{4}$, and the conjugate poles carry $\mp\frac{i}{4}$.

5.3 V is a difference of two arctangents

Because the residues are purely imaginary, the logarithms pair up into pure angles. From $V = \int 2R\, dy = 2\sum_{k=1}^{4} c_k \ln(y - \eta_k)$, group each pole with its conjugate; the η_+ pair is

$$2\Big[\tfrac{i}{4} \ln(y - \eta_+) - \tfrac{i}{4} \ln(y - \bar{\eta}_+)\Big] = \tfrac{i}{2} \ln \frac{y - \eta_+}{y - \bar{\eta}_+}.$$

For real y we have $y-\bar{\eta}_+ = \overline{y-\eta_+}$, so the ratio is $(y-\eta_+)/\overline{(y-\eta_+)} = e^{2i\arg(y-\eta_+)}$, of unit modulus: the $\ln|\cdot|$ parts cancel identically and the pair collapses to $\frac{i}{2}\cdot 2i\arg(y-\eta_+) = -\arg(y-\eta_+)$. The η_- pair (residue $-\frac{i}{4}$) gives $+\arg(y-\eta_-)$ the same way. Hence, using $y-\eta_\pm = (y-a_\pm) - ib_\pm$,

$$\boxed{V(t) = \arg(y-\eta_-) - \arg(y-\eta_+) = \operatorname{atan2}(-b_-,\, y-a_-) - \operatorname{atan2}(-b_+,\, y-a_+), \qquad y = e^t} \tag{5.8}$$

This is manifestly real and, since $b_\pm > 0$, manifestly continuous in t (the arguments never cross the branch cut). *All four* poles are needed for this cancellation: keeping only $\eta_\pm$ leaves a complex V.

5.4 The remaining integral and the real result

$\arg(y-\eta) = \operatorname{Im}\ln(y-\eta)$, and splitting $\ln(y-\eta) = \ln(-\eta) + \ln(1-y/\eta)$ with $\int \ln(1-y/\eta)\,dy/y = -\operatorname{Li}_2(y/\eta)$ gives $\int \ln(y-\eta)\,dy/y = \ln(-\eta)\ln y - \operatorname{Li}_2(y/\eta)$, hence

$$\int \arg(y-\eta)\,dt = \ln y\ \arg(-\eta) - \operatorname{Im}\operatorname{Li}_2(y/\eta). \tag{5.9}$$

In (5.2) we have $t = \ln y$, so $t\,V = \ln y\,[\arg(y-\eta_-) - \arg(y-\eta_+)]$. Combining $\frac{1}{2}[t\,V] - \frac{1}{2}\int V\,dt$ and using $\arg(y-\eta) - \arg(-\eta) = \arg(1-y/\eta)$ on the two $\ln y$ terms, everything assembles into

$$\boxed{J_{\text{arcsinh}} = -\tfrac{1}{2}\Big[\operatorname{Im} G(\eta_+) - \operatorname{Im} G(\eta_-)\Big]_{Y_0}^{Y_1}, \qquad G(\eta) \equiv \operatorname{Li}_2\Big(\frac{y}{\eta}\Big) + \ln y\ \ln\Big(1-\frac{y}{\eta}\Big)} \tag{5.10}$$

with $\operatorname{Im} G(\eta) = \operatorname{Im}\operatorname{Li}_2(y/\eta) + \ln y\ \arg(1-y/\eta)$ and $Y_0 = e^{\operatorname{arcsinh} X_0}$, $Y_1 = e^{\operatorname{arcsinh} X_1}$.

5.5 Bloch–Wigner and Clausen: removing the last complex object

The only non-elementary object left is $\operatorname{Im}\operatorname{Li}_2$. The Bloch–Wigner function

$$\mathcal{D}(z) \equiv \operatorname{Im}\operatorname{Li}_2(z) + \arg(1-z)\ln|z| \tag{5.11}$$

is real-valued and single-valued on $\mathbb{C}\setminus\{0,1\}$. Substituting $\operatorname{Im}\operatorname{Li}_2(z) = \mathcal{D}(z) - \arg(1-z)\ln|z|$ with $z = y/\eta$ and $\ln|z| = \ln y - \ln|\eta|$, the $\ln y$ terms cancel and

$$\operatorname{Im} G(\eta) = \mathcal{D}\Big(\frac{y}{\eta}\Big) + \ln|\eta|\ \arg\Big(1-\frac{y}{\eta}\Big). \tag{5.12}$$

Specializing to (5.5): $\arg\eta_+ = \psi$, $\arg\eta_- = \pi-\psi$, and $|\eta_+| = (w-\nu)/\sqrt{1+m^2} = 1/|\eta_-|$, so

$$\begin{gathered} \psi = \operatorname{atan2}(1,m), \qquad L \equiv \ln|\eta_+| = \ln(w-\nu) - \tfrac{1}{2}\ln(1+m^2) = -\ln|\eta_-|, \\ \varphi_\pm \equiv \arg\Big(1-\frac{y}{\eta_\pm}\Big) = \operatorname{atan2}\big(y,\ w\mp\nu\mp my\big). \end{gathered} \tag{5.13}$$

Finally, Lewin's reduction expresses $\mathcal{D}$ through the Clausen function,

$$\mathcal{D}(re^{i\theta}) = \tfrac{1}{2}\big[\operatorname{Cl}_2(2\theta) + \operatorname{Cl}_2(2\omega) - \operatorname{Cl}_2(2\theta+2\omega)\big], \qquad \omega = -\arg(1-z), \tag{5.14}$$

which holds for all z (Cl_2 is odd and 2π-periodic, so a 2π jump in θ or ω is invisible). With $\theta = \arg(y/\eta_\pm)$ this gives

$$\begin{aligned} \mathcal{D}\Big(\frac{y}{\eta_+}\Big) &= \tfrac{1}{2}\big[\operatorname{Cl}_2(2\psi+2\varphi_+) - \operatorname{Cl}_2(2\psi) - \operatorname{Cl}_2(2\varphi_+)\big], \\ \mathcal{D}\Big(\frac{y}{\eta_-}\Big) &= \tfrac{1}{2}\big[\operatorname{Cl}_2(2\psi) - \operatorname{Cl}_2(2\varphi_-) - \operatorname{Cl}_2(2\psi-2\varphi_-)\big]. \end{aligned} \tag{5.15}$$

Since $\operatorname{Im} G(\eta_+) = \mathcal{D}(y/\eta_+) + L\varphi_+$ and $\operatorname{Im} G(\eta_-) = \mathcal{D}(y/\eta_-) - L\varphi_-$, (5.10) becomes the final, entirely real recipe

$$\boxed{J_{\mathrm{arcsinh}} = -\tfrac{1}{2}\Big[\mathcal{D}\Big(\frac{y}{\eta_+}\Big) - \mathcal{D}\Big(\frac{y}{\eta_-}\Big) + L\,(\varphi_+ + \varphi_-)\Big]_{Y_0}^{Y_1}} \tag{5.16}$$

The y-independent $\mathrm{Cl}_2(2\psi)$ cancels across $Y_0 \to Y_1$, leaving eight Cl_2 evaluations. *No complex arithmetic appears anywhere*: only $\sqrt{\ }$, ln, atan2, and Cl_2.

Evaluating Cl_2. Reduce the argument to $[0,\pi]$ using oddness and 2π-periodicity, then sum

$$\mathrm{Cl}_2(\theta) = -\int_0^\theta \ln\Big|2\sin\frac{u}{2}\Big|du = \sum_{k\geq 1}\frac{\sin k\theta}{k^2} = \theta - \theta\ln\theta + \sum_{k\geq 1} c_k\,\theta^{2k+1}, \qquad c_k = \frac{|B_{2k}|}{2k\,(2k+1)!}, \tag{5.17}$$

with B_{2k} the Bernoulli numbers ($c_1 = \frac{1}{72}$, $c_2 = \frac{1}{14400}, \ldots$). Twenty terms give $\sim 10^{-15}$ uniformly on $[0,\pi]$.

No further collapse. It is tempting to use $\eta_+\eta_- = -1$ and the inversion $\mathcal{D}(1/z) = -\mathcal{D}(z)$ to fold the two Bloch–Wigner terms into one. This fails: the inversion needs $y/\eta_- = 1/(y/\eta_+) = \eta_+/y$, whereas $y/\eta_- = -y\eta_+$ (equal only if $-y^2 = 1$). Numerically $\mathcal{D}(y/\eta_-) \neq \pm\mathcal{D}(y/\eta_+)$, the arguments are not a $z, -z$ duplication pair, and $\varphi_+ + \varphi_-$ is not constant. Equation (5.16) is minimal.

6 The gradient

6.1 Varying the double boundary integral

Use $\hat{n}_A^\alpha\, d\ell_x = \varepsilon_{\alpha\beta} dx^\beta$ (here α, β are Cartesian indices) to rewrite (2.5) without normals:

$$n_A^\alpha d\ell_x\, n_B^\alpha d\ell_y = \varepsilon_{\alpha\beta}dx^\beta \varepsilon_{\alpha\gamma}dy^\gamma = dx^\gamma dy^\gamma \quad\Longrightarrow\quad A_\cap^\sigma = -\oint_{\partial A} dx^\alpha \oint_{\partial B} dy^\alpha\, \Phi_\sigma.$$

Now perturb $\vec{x} \mapsto \vec{x} + \delta\vec{x}$. With $\delta\Phi_\sigma = \frac{\partial \Phi_\sigma}{\partial x^\beta}\delta x^\beta$,

$$\delta A_\cap^\sigma = -\oint_{\partial B} dy^\alpha \oint_{\partial A}\Big[\frac{\partial\Phi_\sigma}{\partial x^\beta}\delta x^\beta dx^\alpha + \Phi_\sigma\, d\delta x^\alpha\Big].$$

Integrating the second term by parts (the boundary term vanishes on a closed loop) and relabelling dummy indices gives an expression antisymmetric in $\alpha\beta$,

$$\delta A_\cap^\sigma = -\oint_{\partial A} dx^\alpha \oint_{\partial B}\Big[dy^\alpha \frac{\partial\Phi_\sigma}{\partial x^\beta} - dy^\beta \frac{\partial \Phi_\sigma}{\partial x^\alpha}\Big]\delta x^\beta,$$

and restoring $\varepsilon_{\alpha\beta}dy^\beta = \hat{n}_B^\alpha d\ell_y$ collapses it, via (2.4) and $\vec{E}_\sigma = \nabla\Phi_\sigma$, to the remarkably clean

$$\boxed{\delta A_\cap^\sigma = -\oint_{\partial A} dx^\alpha\,\delta x^\beta\,\varepsilon_{\alpha\beta}\,\Psi_B^\sigma(\vec{x})} \tag{6.1}$$

The entire variation is weighted by the softened membership Ψ_B^σ — the mollified analogue of "how much of B is here." No derivative of an intersection point is ever needed.

6.2 Vertex gradient

Traverse the edges of ∂A with $\vec{x}_k(s) = \vec{v}_k + s\vec{e}_k = (1-s)\vec{v}_k + s\vec{v}_{z(k)}$, so $\delta\vec{x}_k = (1-s)\delta\vec{v}_k + s\,\delta\vec{v}_{z(k)}$ and $dx_k^\alpha = e_k^\alpha ds$. Defining the two moments

$$W_k^0 \equiv \int_0^1 (1-s)\,\Psi_B^\sigma(\vec{x}_k(s))\,ds, \qquad W_k^1 \equiv \int_0^1 s\,\Psi_B^\sigma(\vec{x}_k(s))\,ds, \tag{6.2}$$

and collecting the coefficient of each $\delta\vec{v}_m$,

$$\boxed{\frac{\partial A_\cap^\sigma}{\partial v_m^\beta} = \varepsilon_{\beta\alpha}\Big[W_m^0\,e_m^\alpha + W_{z'(m)}^1\,e_{z'(m)}^\alpha\Big]} \tag{6.3}$$

(z' = previous vertex): each edge deposits W^0 of its cross term on its start vertex and W^1 on its end vertex.

6.3 Ψ_B^σ in closed form

With $\hat{n}_B^\alpha d\ell_y = \varepsilon_{\alpha\beta}dy^\beta$ and $\vec{y}_k(s) - \vec{x} = -\vec{w}_k + \vec{e}_k s$ ($\vec{w}_k = \vec{x} - \vec{v}_k$), (2.4) for one edge is

$$\Psi_k^\sigma(\vec{x}) = \int_0^1 ds\,\frac{-w_k^\alpha + e_k^\alpha s}{2\pi(|\vec{w}_k - \vec{e}_k s|^2 + \sigma^2)}\,\varepsilon_{\alpha\beta}e_k^\beta.$$

The $\vec{e}_k s$ term dies ($\vec{e}_k \times \vec{e}_k = 0$), leaving a single arctangent: with $a_k = |\vec{e}_k|^2$, $b_k = -2\vec{w}_k\cdot\vec{e}_k$, $c_k = |\vec{w}_k|^2 + \sigma^2$ and $D_k = \sqrt{4a_kc_k - b_k^2}$,

$$\Psi_k^\sigma(\vec{x}) = -\frac{w_k^\alpha\varepsilon_{\alpha\beta}e_k^\beta}{\pi D_k}\Big[\arctan\frac{2a_k + b_k}{D_k} - \arctan\frac{b_k}{D_k}\Big], \qquad \Psi_B^\sigma = \sum_{k\in\partial B}\Psi_k^\sigma. \tag{6.4}$$

6.4 The moments via the master form

Equation (6.2) is an outer integral over an A-edge of the arctangent (6.4). Resolving the field point along and across the inner edge $\hat{e}_k = \vec{e}_k/|\vec{e}_k|$ — write P_0, P_1 for the along components and X_0, X_1 for the across components of $(\vec{v}_m - \vec{v}_k)$ and $\vec{e}_m$ — the across coordinate is $\xi = X_0 + sX_1$ and, because $D_k = 2|\vec{e}_k|\sqrt{\xi^2+\sigma^2}$ and $\vec{w}_k\times\vec{e}_k = |\vec{e}_k|\xi$, the prefactor and the arctangent arguments reduce to exactly the Θ of (3.6). Changing variable from s to ξ, the two moments are the ξ-integrals of $\frac{\xi}{\sqrt{\xi^2+\sigma^2}}\Theta$ and $\frac{\xi^2}{\sqrt{\xi^2+\sigma^2}}\Theta$ — i.e. precisely rows two and three of the master table (4.6):

$$W^0 \;\longleftrightarrow\; M_1 \equiv \mathbb{M}\big[\sqrt{\xi^2+\sigma^2}\big] = \sqrt{\xi^2+\sigma^2}\,\Theta - \lambda, \qquad W^1 \;\longleftrightarrow\; M_1' \equiv \mathbb{M}[v_-] = v_-\Theta - \tfrac{1}{2}\varrho - J_{\text{arcsinh}}. \tag{6.5}$$

So the force needs *no new analysis whatsoever*: W^0 is elementary, and W^1's only transcendental is the same J_{arcsinh} that closes the energy — reached through v_- rather than the energy's v_+, and entering with the opposite sign. One Clausen evaluation serves both.

7 The near-parallel bridge

The assembled panel and moments divide by $X_1 = \vec{e}_m \times \hat{e}_k = |\vec{e}_m|\sin\theta_{mk}$, which vanishes as two edges align: the energy carries $1/X_1$, the W^1 moment $1/X_1^2$, and the slope $\mu = P_1/X_1 \to \infty$. The closed forms then lose all precision to cancellation — at $|\sin\theta| \sim 10^{-6}$ the panel is wrong in the sixth digit, and finite-differencing the energy amplifies that by $1/2h$ into an $O(10^{-2})$ force error.

The cure is to recognize that each of these quantities is a *perfectly regular integral*; only the closed form is singular. Undoing the X_1 division with $\xi = X_0 + uX_1$, $u \in [0,1]$:

$$\int_0^1 \sqrt{\xi^2+\sigma^2}\,\Theta\,du, \qquad \int_0^1 \frac{\xi\,\Theta}{\sqrt{\xi^2+\sigma^2}}\,du, \qquad \int_0^1 \frac{u\,\xi\,\Theta}{\sqrt{\xi^2+\sigma^2}}\,du, \qquad \Theta = \arctan\frac{U_0+P_1u}{\sqrt{\xi^2+\sigma^2}}, \tag{7.1}$$

all manifestly finite as $X_1 \to 0$, with limits equal to the exactly-parallel branch. The only feature is the arctangent's node $u_\star = -U_0/P_1$, a peak of width $\sim \sigma$; splitting the interval there and applying a fixed Gauss rule (2×24 nodes) holds $\sim 10^{-15}$ down to $\sigma \sim 10^{-3}$. Used for $|X_1| \le 10^{-2}|\vec{e}_m|$ and the closed forms elsewhere, this restores machine precision uniformly.

8 Packing energy and exact Newton minimization

8.1 The energy

For a packing of N polygons the overlap energy is the normalized, squared form

$$U_{\text{ov}} = 2\sum_{A<B}\left(\frac{A_\cap^{\sigma,AB}}{A_A^t + A_B^t}\right)^2, \tag{8.1}$$

where A^t are the constant *target* areas. Squaring makes the contact force vanish smoothly as an overlap closes, so the packing jams near the target packing fraction instead of collapsing; the constant normalizer keeps $\partial U/\partial\vec{v} = 4\sum(A_\cap/\text{norm}^2)\,\partial A_\cap/\partial\vec{v}$, so (6.3) is all that is needed. To this we add the soft-body springs (area and perimeter, in relative form) and an intra-polygon self-repulsion barrier. Far pairs are pruned with a neighbour list and a C^2 switch on the centroid separation, so excluded pairs contribute *exactly* zero and the energy stays smooth.

8.2 Why Newton is now available

Newton's method needs two things that the sharp model cannot supply:

1. a C^∞ energy, so the Hessian exists — provided by mollification;

2. a force that is *self-consistent* with its own energy to near machine precision, i.e. $\vec{F} = -\nabla U$ to far better than the step size — provided by the closed forms above, once the near-parallel bridge (§7) removes the cancellation.

Point 2 is the subtle one: a force that is merely *accurate* is not enough. If $\vec{F}$ and ∇U disagree by ε, the finite-difference Hessian inherits ε/h and Newton stalls at that level. Before the bridge, the exact tier disagreed with itself by $\sim 10^{-2}$ and Newton stalled there; after it, the agreement is $\sim 10^{-9}$ (the finite-difference floor) and Newton runs to the numerical bottom.

8.3 The scheme

1. **FIRE** from the initial configuration to reach the basin, to $\max|F| \sim 10^{-6}$. FIRE is robust and cheap but first-order, so it cannot go much further.

2. **Newton polish**: solve $H\,\delta\vec{v} = -\vec{F}$ and step, with H the Hessian obtained by central-differencing the analytic force, $2\times(2N_v)$ force evaluations per step.

At $N = 6$ this converges in four steps with textbook quadratic behaviour,

$$9.3 \times 10^{-7} \longrightarrow 3.7 \times 10^{-8} \longrightarrow 1.7 \times 10^{-12} \longrightarrow \max|F| = 9.3 \times 10^{-13},$$

about nine decades below the sharp model's 3.6×10^{-3} floor. That gap is the whole point of mollifying.

8.4 Cost

The finite-difference Hessian is the bottleneck: $2 \times (2N_v)$ analytic force evaluations per Newton step, fine at $N = 6$ but impractical by $N = 32$. The remedy is the *analytic* Hessian — equivalently the stiffness matrix, so it is wanted anyway — derived in §9. It costs one pass over edge pairs, comparable to a single force evaluation, and introduces *no* new transcendentals.

9 The analytic Hessian

We now differentiate (6.3) once more. The pleasant surprise is that the second derivative is *easier* than the first: its one new ingredient is the gradient of the softened membership, which turns out to be a boundary integral of the *kernel itself* and is entirely elementary.

9.1 What has to be differentiated

Write the force as $g_m^\beta \equiv \partial A_\cap^\sigma / \partial v_m^\beta = \varepsilon_{\beta\alpha}\big[W_m^0 e_m^\alpha + W_{z'(m)}^1 e_{z'(m)}^\alpha\big]$. Differentiating with respect to v_n^γ produces exactly two kinds of term:

$$H_{mn}^{\beta\gamma} \equiv \frac{\partial^2 A_\cap^\sigma}{\partial v_m^\beta\, \partial v_n^\gamma} = \underbrace{\varepsilon_{\beta\alpha}\Big[\frac{\partial W_m^0}{\partial v_n^\gamma} e_m^\alpha + \frac{\partial W_{z'(m)}^1}{\partial v_n^\gamma} e_{z'(m)}^\alpha\Big]}_{\text{moment part}} + \underbrace{\varepsilon_{\beta\gamma}\Big[W_m^0\big(\delta_{z(m)n} - \delta_{mn}\big) + W_{z'(m)}^1\big(\delta_{mn} - \delta_{z'(m)n}\big)\Big]}_{\text{geometric part}}, \tag{9.1}$$

using $\partial e_k^\alpha / \partial v_n^\gamma = \delta^{\alpha\gamma}(\delta_{z(k)n} - \delta_{kn})$ and $z(z'(m)) = m$. The geometric part is free: it reuses the W^0, W^1 already computed for the force, weighted by Kronecker deltas. All the work is in $\partial W / \partial v$.

9.2 The one new object: $\nabla \Psi_B^\sigma$

Since $W_k^{0,1}$ are integrals of Ψ_B^σ along an edge of A, moving a vertex of A moves the *field point*, and the chain rule needs $\nabla_{\vec{x}} \Psi_B^\sigma$. Using $\Psi_B^\sigma = K_\sigma * \mathbf{1}_B$ and $\nabla_{\vec{x}} K_\sigma(\vec{x} - \vec{y}) = -\nabla_{\vec{y}} K_\sigma(\vec{x} - \vec{y})$, the gradient theorem on B gives

$$\boxed{\nabla_{\vec{x}} \Psi_B^\sigma(\vec{x}) = -\oint_{\partial B} K_\sigma(|\vec{x} - \vec{y}|)\, \hat{n}_B(\vec{y})\, d\ell_y} \tag{9.2}$$

The potential Φ_σ and the field $\vec{E}_\sigma$ have dropped out: differentiating the softened membership returns the *mollifier itself*, smeared along ∂B. This is the analogue, one derivative up, of $\Psi_B^\sigma = \oint \vec{E}_\sigma \cdot \hat{n}_B$, and it is why the Hessian needs no new special functions.

Closed form per edge. On edge k of ∂B, with $\vec{w}_k = \vec{x} - \vec{v}_k$ and the same $a_k = |\vec{e}_k|^2$, $b_k = -2\vec{w}_k\vec{e}_k$, $c_k = |\vec{w}_k|^2 + \sigma^2$, $D_k^2 = 4a_kc_k - b_k^2$ as in (6.4), we have $|\vec{x} - \vec{y}_k(r)|^2 + \sigma^2 = a_kr^2 + b_kr + c_k \equiv q_k(r)$, so

$$\nabla_{\vec{x}}\Psi_B^\sigma = -\sum_{k\in\partial B} \hat{n}_k\,|\vec{e}_k|\,\frac{\sigma^2}{\pi}\int_0^1 \frac{dr}{q_k(r)^2}, \tag{9.3}$$

and the reduction $\int dr/q^2 = \frac{2ar+b}{D^2q} + \frac{2a}{D^2}\int \frac{dr}{q}$ together with $\int dr/q = \frac{2}{D}\arctan\frac{2ar+b}{D}$ gives the elementary closed form

$$\int_0^1 \frac{dr}{q^2} = \left[\frac{2ar+b}{D^2\,q(r)}\right]_0^1 + \frac{4a}{D^3}\left[\arctan\frac{2ar+b}{D}\right]_0^1. \tag{9.4}$$

Rational plus arctangent — nothing more.

9.3 Moving B: the cross block

If instead a vertex of B moves, the boundary ∂B itself moves and $\delta\Psi_B^\sigma = \oint_{\partial B} K_\sigma(|\vec{x}-\vec{y}|)\,(\delta\vec{y}\hat{n}_B)\,d\ell_y$. With $\delta\vec{y} = (1-r)\delta\vec{v}_k + r\,\delta\vec{v}_{z(k)}$ on edge k,

$$\frac{\partial\Psi_B^\sigma}{\partial v_n^\gamma} = \sum_{k\in\partial B} \hat{n}_k^\gamma\,|\vec{e}_k|\,\frac{\sigma^2}{\pi}\Big[\delta_{kn}\int_0^1 \frac{(1-r)\,dr}{q_k^2} + \delta_{z(k)n}\int_0^1 \frac{r\,dr}{q_k^2}\Big], \tag{9.5}$$

which needs only the first moment $\int r\,dr/q^2 = -\frac{1}{2a}\big[q^{-1}\big] - \frac{b}{2a}\int dr/q^2$, again elementary. So the A–B cross block costs the same objects as (9.4).

9.4 Second moments and the extended master form

Moving a vertex of A on edge k shifts the field point by $\partial\vec{x}_k(s)/\partial v_n = \big[(1-s)\delta_{kn} + s\,\delta_{z(k)n}\big]\mathbb{1}$, so with the *second moments*

$$\vec{S}_k^{(pq)} \equiv \int_0^1 (1-s)^p\,s^q\,\nabla\Psi_B^\sigma\big(\vec{x}_k(s)\big)\,ds, \qquad (p,q)\in\{(2,0),(1,1),(0,2)\}, \tag{9.6}$$

the chain rule gives compactly

$$\frac{\partial W_k^0}{\partial v_n^\gamma} = S_k^{(20)\gamma}\delta_{kn} + S_k^{(11)\gamma}\delta_{z(k)n}, \qquad \frac{\partial W_k^1}{\partial v_n^\gamma} = S_k^{(11)\gamma}\delta_{kn} + S_k^{(02)\gamma}\delta_{z(k)n}. \tag{9.7}$$

It remains to do the outer s-integral in (9.6). Substituting (9.3) and passing to the normalized frame of (3.2), the rational part of (9.4) integrates to a rational function of X, while the arctangent part carries the prefactor $4a/D^3$ with $D = 2|\vec{e}|\sigma\sqrt{X^2+1}$ — that is, weight $(X^2+1)^{-3/2}$ against the same Θ of (3.6). So the Hessian is a fourth instance of the master form (4.1), with $V' = (X^2+1)^{-3/2}$, hence $V = X/\sqrt{X^2+1}$:

$$\mathbb{M}\big[X/\sqrt{X^2+1}\big] = \frac{X}{\sqrt{X^2+1}}\,\Theta - \int \frac{X(m-\nu X)}{(X^2+1)\,F(X)}\,dX. \tag{9.8}$$

The integrand of the correction is *rational*: the weight's $\sqrt{X^2+1}$ cancels the one in Θ' (4.2) outright. Partial-fractioning over the two irreducible quadratics X^2+1 and F leaves logarithms and arctangents only. Extending (4.6):

$$
\begin{array}{lllll}
\text{weight } V' & V & \int V\Theta' & \text{transcendental} & \text{used by} \\
\sqrt{X^2+1} & v_+ & \frac{1}{2}\varrho - J_{\text{arcsinh}} & +J_{\text{arcsinh}} & \text{energy} \\
X/\sqrt{X^2+1} & \sqrt{X^2+1} & \lambda & \text{none} & \text{force } W^0 \\
X^2/\sqrt{X^2+1} & v_- & \frac{1}{2}\varrho + J_{\text{arcsinh}} & -J_{\text{arcsinh}} & \text{force } W^1 \\
(X^2+1)^{-3/2} & X/\sqrt{X^2+1} & \text{rational} & \text{none} & \text{Hessian}
\end{array}
\tag{9.9}
$$

9.5 Assembly and remarks

Substituting (9.7) into (9.1) gives the A–A block

$$
\begin{aligned}
H^{\beta\gamma}_{mn} = \; & \varepsilon_{\beta\alpha}\Big[\big(S^{(20)\gamma}_{m}\delta_{mn} + S^{(11)\gamma}_{m}\delta_{z(m)n}\big)e^{\alpha}_{m} + \big(S^{(02)\gamma}_{z'(m)}\delta_{mn} + S^{(11)\gamma}_{z'(m)}\delta_{z'(m)n}\big)e^{\alpha}_{z'(m)}\Big] \\
& + \varepsilon_{\beta\gamma}\Big[W^0_m\big(\delta_{z(m)n} - \delta_{mn}\big) + W^1_{z'(m)}\big(\delta_{mn} - \delta_{z'(m)n}\big)\Big],
\end{aligned}
\tag{9.10}
$$

and the A–B cross block follows by replacing $\nabla\Psi^{\sigma}_{B}$ with (9.5) inside (9.6). Three observations:

- **No new transcendentals.** The Hessian reuses W^0, W^1 (which carry J_{arcsinh} from the force) and adds only *elementary* second moments. The dilogarithm never reappears; in fact the new row of (9.9) is the mildest of the four.
- **Sparsity.** Each vertex couples only to vertices of its own two edges and to the vertices of neighbouring polygons that survive the neighbour list, so H is sparse and banded up to the pair list — exactly the structure a stiffness matrix should have.
- **Symmetry.** $H^{\beta\gamma}_{mn} = H^{\gamma\beta}_{nm}$ follows from the $A \leftrightarrow B$ symmetry of (2.5); it is a useful implementation check, since the two blocks are assembled by different code paths.
- **Cost.** One pass over edge pairs, i.e. the same order as a single force evaluation, versus $2\times(2N_v)$ force evaluations for the finite-difference Hessian — a factor $\sim 4N_v$ (about $250\times$ at $N = 32$).

The near-parallel caveat of §7 applies here too: $\vec{S}^{(pq)}$ inherits the $1/X_1$ powers, and the same regular-u-integral bridge should be used inside the band.

10 Implementation

10.1 From the pair Hessian to the packing Hessian

Section 9 gives $\partial^2 A^{\sigma}_{\cap}/\partial\vec{v}\,\partial\vec{v}$ for one polygon pair. The energy actually minimized is the normalized, squared form (8.1), and each pair carries the smooth switch, so one more chain rule is needed. Writing $a = \sum_{\text{images}} S(d)\,A_{\cap}$ for the switched, image-summed overlap of a pair and $\text{nrm} = A^t_A + A^t_B$,

$$
\delta a = \textstyle\sum_{\text{im}}\big[S\,\delta A_{\cap} + A_{\cap}S'\,\delta d\big], \tag{10.1}
$$

$$
\delta^2 a = \textstyle\sum_{\text{im}}\big[S\,\delta^2 A_{\cap} + S'\big(\delta d\otimes\delta A_{\cap} + \delta A_{\cap}\otimes\delta d\big) + A_{\cap}\big(S''\,\delta d\otimes\delta d + S'\,\delta^2 d\big)\big], \tag{10.2}
$$

and then

$$
\frac{\partial U}{\partial\vec{v}} = \frac{4a}{\text{nrm}^2}\,\delta a, \qquad \frac{\partial^2 U}{\partial\vec{v}\,\partial\vec{v}} = \frac{4}{\text{nrm}^2}\Big(\delta a\otimes\delta a + a\,\delta^2 a\Big). \tag{10.3}
$$

The switch depends on the vertices only through the centroid separation $d = |\vec{c}_B + \text{shift} - \vec{c}_A|$, so with $\hat{d}$ the unit separation, $\varepsilon_n = -1$ for $n \in A$ and $+1$ for $n \in B$, and n_v the vertex count of the polygon owning n,

$$\frac{\partial d}{\partial \vec{v}_n} = \frac{\varepsilon_n \hat{d}}{n_v}, \qquad \frac{\partial^2 d}{\partial \vec{v}_n \partial \vec{v}_m} = \frac{\varepsilon_n \varepsilon_m}{n_v^{(n)} n_v^{(m)}} \frac{\mathbb{1} - \hat{d}\hat{d}^\mathsf{T}}{d}. \tag{10.4}$$

Note the $\delta a \otimes \delta a$ term in (10.3): because the energy is *squared*, the Hessian gets a rank-one piece built from the pair gradient even where $\delta^2 A_\cap$ vanishes.

10.2 Exact inner, quadratured outer

The implementation evaluates the inner ∂B integrals in closed form — Ψ_B^σ by (6.4), $\nabla \Psi_B^\sigma$ by (9.4), and $\partial \Psi_B^\sigma / \partial \vec{v}_B$ by (9.5) — and takes the *outer* edge integral (9.6) by Gauss quadrature. This is deliberate. Newton's converged configuration is determined by where the *force* vanishes, not by the Hessian: an inexact H changes the path and the convergence rate, but the fixed point is $\vec{F} = \vec{0}$, which is evaluated with the exact tier of §3–§5. So outer quadrature in the Hessian costs nothing in final precision, while removing the entire $2 \times (2N_v)$-force-evaluation cost of a finite-difference Hessian. (The fully closed-form outer integral is available if wanted — it is row four of (9.9) — but buys only speed.)

10.3 Code map

Symbol / step	`energies.py`
$I_0 = \int_0^1 dr/q^2$, $I_1 = \int_0^1 r\, dr/q^2$ (9.4)	`_plummerQuadInts`
Ψ_B^σ (6.4)	`plummerMeasure`
$\nabla_{\vec{x}} \Psi_B^\sigma$ (9.2)	`_gradPlummerMeasure`
$\partial \Psi_B^\sigma / \partial \vec{v}_B$ (9.5)	`_dPlummerMeasureDvB`
moments $W^{0,1}$, $\vec{S}^{(pq)}$, blocks (9.10)	`_pairHessianOneSided`
pair Hessian (all four blocks)	`plummerPairHessian`
packing Hessian (10.1)–(10.4)	`plummerOverlapHessian`
Newton hook	`minimize.minimizeNewton(hessian=...)`

Quadrature order matters more here than for the force. The outer rule for the Hessian must be finer than the one used for Ψ_B^σ in the force, because $\nabla \Psi_B^\sigma$ is sharper than Ψ_B^σ (it is the kernel, not its potential). Reusing the force's 32-point rule leaves the Hessian only $\sim 8 \times 10^{-7}$ accurate and Newton loses its quadratic tail entirely — it crawls, taking 30 steps and stalling near 3×10^{-12}. A dedicated 96-point rule (`_HNODES`) restores it: Newton then converges in *two* steps. The remaining $\sim 2.5 \times 10^{-7}$ is not quadrature at all — it is the gap between the semi-analytic and exact tiers, i.e. the exact tier's own $\sim 3 \times 10^{-9}$ gradient noise amplified by the $1/\sigma$ of differentiation; refining the rule past 64 points changes nothing.

The B–B block is obtained by calling `_pairHessianOneSided` with the loops swapped, and the two cross blocks are computed independently; that they agree to machine precision ($|H - H^\mathsf{T}| = 0$ exactly in the assembled pair matrix) is a strong check, since they come from different code paths — one differentiates the field point, the other moves ∂B.

The soft-body springs and the self-repulsion barrier are cheap and local; their Hessian is still taken by finite differences, which costs $2 \times (2N_v)$ evaluations of a *cheap* force and leaves the expensive mollified-overlap part fully analytic.

11 Numerical verification

Every step above is checked against an independent computation; all agreements are reported as maximum absolute error over randomized parameters.

Quantity	Checked against	Agreement
Panel (3.5)	direct t-quadrature	$\sim 2 \times 10^{-14}$
Assembly (3.7)	direct (t, s)-quadrature	$\sim 2 \times 10^{-13}$
Λ_0, Λ_1 (3.8)–(3.9)	direct quadrature	$\sim 3 \times 10^{-13}$
λ, ϱ (4.7)–(4.8)	direct quadrature	$\sim 10^{-15}$
Master table (4.6)	quadrature, each weight	$\sim 10^{-16}$
Poles (5.5), residues (5.7)	`roots`(D), N/D'	$\sim 10^{-15}$
V (5.8)	finite difference of (5.1) integrand	$\sim 10^{-10}$
J_{arcsinh} (5.10), (5.16)	direct integral (4.5)	$\sim 10^{-16}$
Cl_2 series (5.17)	`mpmath`	$\sim 10^{-15}$
Gradient (6.3)	central difference of $A^{\sigma}_{\cap}$	$\sim 3 \times 10^{-9}$
$\nabla\Psi^{\sigma}_{B}$ (9.2)	central difference of Ψ^{σ}_{B}	$\sim 3 \times 10^{-10}$
$\int dr/q^2$ (9.4)	direct quadrature	$\sim 2 \times 10^{-15}$
Hessian master row (9.8)	direct quadrature	$\sim 6 \times 10^{-16}$
Near-parallel bridge (7.1)	dense 2D reference	$\sim 10^{-16}$
Exact-tier self-consistency	semi-analytic tier	$\sim 3 \times 10^{-9}$
Pair Hessian (9.10)	central difference of the pair gradient	$\sim 5 \times 10^{-11}$
Pair Hessian symmetry	$\lvert H - H^{\mathsf{T}}\rvert$ (blocks from distinct paths)	0 exactly
Packing Hessian (10.3)	central difference of the exact gradient	$\sim 2.5 \times 10^{-7}$*

*at the tier floor, not an implementation error: the semi/exact gradient gap (3×10^{-9}) divided by σ reproduces it, and the analytic Hessian is *closer* to the exact-tier Hessian (2.5×10^{-7}) than the semi-tier Hessian is (7.8×10^{-7}).

Newton at $N = 6$, same start ($\max\lvert F\rvert = 9.3 \times 10^{-7}$)	steps	wall clock
finite-difference Hessian, $\max\lvert F\rvert = 9.3 \times 10^{-13}$	4	345 s
analytic Hessian, $\max\lvert F\rvert = 6.2 \times 10^{-13}$	**2**	**6 s**

The analytic Hessian builds in 0.9 s against 83 s for the finite-difference one ($\sim 110\times$), giving $\sim 57\times$ end to end — and the advantage grows linearly with N_v, since the finite-difference cost is $2(2N_v)$ *force* evaluations while the analytic cost is one edge-pair pass.

Supporting scripts: `notes/verifyPanelFrame.py` (the normalized frame, (3.5) and (3.7)), `notes/master_form.py` (master table), `notes/verify_realroute.py` (V, J_{arcsinh}, the no-collapse check), `notes/partial_fractions_arcsinh.py` (poles, residues, Clausen recipe), `notes/verify_gradient_masters.py` (M_1, M_1'), `notes/arcsinhClausen.nb` (Clausen coefficients and cross-checks). The implementation lives in `energies.py` (`_masterM`, `_tCoreReal`, `_cl2`, `_ceBridge`, `_wBridge`).

12 Summary

The mollified overlap of two polygons reduces, exactly, to a finite sum over edge pairs of elementary functions plus one Clausen function:

- mollification turns a 2D area integral into a double *boundary* integral, (2.5), because the Plummer kernel's potential is a logarithm;
- each edge-pair panel splits into logarithmic pieces (fully elementary, (3.8)) and arctangent pieces;
- all arctangent pieces — for both the energy and the force — are instances of one master integral $\mathbb{M}[V]$, (4.6), whose only transcendental is J_{arcsinh};
- J_{arcsinh} evaluates in closed form with no complex arithmetic, (5.16), via two poles in closed form, purely imaginary residues, a two-arctangent antiderivative, and Bloch–Wigner $\to \mathrm{Cl}_2$;
- the gradient follows from the same machinery with Ψ_B^σ as the weight, (6.1)–(6.3), and shares the identical J_{arcsinh};
- with the near-parallel bridge in place the force is self-consistent to $\sim 10^{-9}$, which is what makes Newton viable — driving $\max |F|$ to $\sim 10^{-12}$, nine decades past the sharp model;
- the analytic Hessian, (9.10), needs one genuinely new object, $\nabla \Psi_B^\sigma = -\oint K_\sigma \hat{n}_B$, (9.2) — the mollifier smeared along ∂B — which is elementary, so the second derivative introduces *no* new transcendentals and costs one edge-pair pass instead of $2(2N_v)$ force evaluations.